ATTENDEASE · SOFTWARE ENGINEERING MINI PROJECT

Branch Coverage Report

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Build under test	`main` @ `92bcd23` (merge of PR #113, 1 Nov 2026)
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Decision (branch) coverage of the AttendEase backend, measured with coverage.py.

1. Summary

92.3%	155/168	9	91.7%
SERVICE-LAYER BRANCHES	BRANCH OUTCOMES TAKEN	PARTIALLY COVERED DECISIONS	ALL APPLICATION CODE

Branch coverage measures whether each possible outcome of every decision point (each `if`, loop, `try/except` and conditional expression) was taken at least once. In the service layer the test suite takes **155 of 168** branch outcomes (**92.3%**). **9** decision points have one outcome that is never taken. The other 4 untaken outcomes belong to two decisions inside `except IntegrityError` race handlers that never run in the tests (see section 5). Across all application code, branch coverage is **91.7%** (222 of 242).

Verdict (TC-12, NFR-006): PASS Branch coverage of the service layer is above the 70% target.

2. Method

Item	Detail
Operating system	Linux x86-64 (2 vCPU)
Python	3.13.17 (local run) · 3.11 in GitHub Actions CI
Framework	Django 4.2.30, Django REST Framework 3.17.2, SimpleJWT 5.5.1
Database	SQLite (Django test database, created and destroyed per run)
Test runner	Django `manage.py test` (unittest) with unittest-xml-reporting for per-test results
Coverage tool	coverage.py 7.17.0, branch measurement enabled
Static checks	Ruff 0.17.1: all checks passed · `manage.py check`: no issues
Measurement run	`coverage run --branch --source=users,academic,attendance,corrections,analytics manage.py test` (320 tests, all passed)
Service layer (NFR-006 scope)	The 7 files enforced by CI: `users/services.py`, `users/security.py`, `academic/services.py`, `attendance/services.py`, `corrections/services.py`, `corrections/notifications.py`, `analytics/services.py`
Application code	All source in the five apps, excluding tests, migrations and empty `__init__.py` files
CI gate	`coverage report --include=<service files> --fail-under=70` on every pull request

A branch is written as *source line → destination*. "Function exit" means that outcome leaves the function, for example an early `return` that was never reached.

3. Service layer by file

File	Responsibility	Branches	Taken	Missed	Partial	Coverage	
`academic/services.py`	Courses, sections, enrollment, timetable, CSV, assignment	50	45	5	5	**90.0%**	
`analytics/services.py`	Percentages, shortage, reports	18	18	0	0	**100.0%**	
`attendance/services.py`	Sessions, marking, edit window, audit	62	58	4	2	**93.5%**	
`corrections/notifications.py`	Faculty email notification	8	8	0	0	**100.0%**	
`corrections/services.py`	Submit, queue, approve, reject	22	20	2	0	**90.9%**	
`users/security.py`	Login lockout (SEC-002)	2	1	1	1	**50.0%**	
`users/services.py`	Registration, login, logout, password reset	6	5	1	1	**83.3%**	
Service layer total		**168**	**155**	**13**	**9**	**92.3%**	

4. All application code by app

App	Branches	Taken	Coverage	
users	34	27	**79.4%**	
academic	64	59	**92.2%**	
attendance	84	78	**92.9%**	
corrections	34	32	**94.1%**	
analytics	26	26	**100.0%**	
Total	**242**	**222**	**91.7%**	

5. Branch outcomes not taken (service layer)

Each row is one outcome that the tests never took. *Partial*: the decision ran and its other outcome was tested. *Not reached*: the decision is inside a race-condition handler that never runs in the tests.

File	Branch	Kind	Function	Decision
academic/services.py	54 → line 56	Partial	update_course	`if Course.objects.filter(code=code).exclude(pk=course.pk).exists():`
academic/services.py	161 → line 162	Partial	add_timetable_slot	`if end_time <= start_time:`
academic/services.py	200 → line 202	Partial	sections_for_user	`if user.role == User.Role.STUDENT:`
academic/services.py	218 → line 219	Partial	_read_csv_emails	`if upload.size > MAX_CSV_BYTES:`
academic/services.py	239 → line 240	Partial	_read_csv_emails	`if len(rows) > MAX_CSV_ROWS:`
attendance/services.py	165 → line 166	Not reached	submit_records	`if existing.exists():`
attendance/services.py	165 → line 167	Not reached	submit_records	`if existing.exists():`
attendance/services.py	201 → line 203	Partial	sessions_for_user	`if user.role == User.Role.FACULTY:`
attendance/services.py	444 → line 450	Partial	edit_closed_record	`if old_status != status.upper():`
corrections/services.py	65 → line 66	Not reached	submit_correction	`if pending.exists():`
corrections/services.py	65 → line 67	Not reached	submit_correction	`if pending.exists():`
users/security.py	28 → line 29	Partial	record_failed_login	`if attempt.expires_at <= now:`
users/services.py	76 → line 77	Partial	register_user	`if role not in (User.Role.STUDENT, User.Role.FACULTY):`

6. Combined figure and conclusion

Measure (service layer)	Covered	Total	Coverage
Statements	542	577	93.9%
Branches	155	168	92.3%
Combined (reported by CI)	**697**	**745**	**93.6%**

Both statement and branch coverage of the service layer exceed the 70% target of NFR-006 / TC-12. The outcomes that were not taken are defensive checks, listed in section 5, and are candidates for further tests if the suite is extended.